Jiaze Li

jiazeli0329.github.io

Email : jiaze.li@ucdconnect.ie
Mobile : +86-188-3366-1166

Education

University College Dublin, Dublin, Ireland *Sept. 2022 – Jul. 2026*
B.E. in Electronic Information Engineering. Cumulative GPA: 3.36/4.20

Beijing University of Technology, Beijing, China *Sept. 2022 – Jul. 2026*
B.E. in Electronic Information Engineering. Cumulative GPA: 3.33/4.20

Research Experience

Event-Guided Diffusion Transformer for Video Frame Interpolation [Undergraduate Thesis]
Supervised by Prof. Yongjian Deng at Data Mining and Security Lab. *Oct. 2025 – Jul. 2026*

- Developed an event-guided Diffusion Transformer for video frame interpolation, integrating asynchronous event streams into a DiT backbone via cross-attention to resolve severe motion blur.
- Designed a motion-aware supervised learning strategy with representation alignment, dynamically regularizing learning objectives across complex non-uniform motion regions to enforce spatial-temporal coherence.
- Outperformed baselines on Middlebury and SNU-FILM benchmarks, reducing LPIPS error by 21.4% and boosting FloLPIPS temporal consistency by 27.3%.

Unified Vision-Language Model for Photovoltaic Remote Sensing
Supervised by Prof. Zhiling Guo and Prof. Jinyue Yan at DigiEnergy Lab. *Jul. 2025 – May 2026*

- Developed a PV-specific Vision-Language Model based on a curated 1M-pair dataset to enable zero-shot and few-shot learning for downstream photovoltaic interpretation tasks.
- Designed Knowledge Assistance Module featuring gated-convolutional attention maps and cross-attention fusion to adaptively inject pre-trained spatial priors into baseline encoders.
- Achieved SOTA performance across downstream tasks, reaching 97.96% classification accuracy, a +19.35% segmentation IoU gain, a +16.7% mAP_{50} detection boost, and a +6.56% Δ_{CLIP} score in image generation.

Generative Foreign Object Detection in High-Speed Railway
Supervised by Prof. Liguo Zhang at Cyber-Physical Network System Lab. *Apr. 2024 – Dec. 2025*

- Developed a generation-extraction-fusion pipeline with LoRA-fine-tuned Stable Diffusion to synthesize high-fidelity foreign object samples, resolving data scarcity and annotation bottlenecks in high-speed railway.
- Enhanced YOLOv9 by integrating Star-Operation backbone and A-DyS dynamic sampling module to effectively capture subtle details of distant small objects amidst complex background noise.
- Achieved mAP_{50} of 95.8% (+5.4%) and mAP_{50-95} of 61.2% (+6.4%) via synthetic-augmented training, drastically reducing missed detections.

Synthetic-Driven Multimodal AI for PV Segmentation
Supervised by Prof. Zhiling Guo at DigiEnergy Lab. *Feb. 2024 – Nov. 2025*

- Built a text-to-image framework to synthesize paired remote sensing images and pixel-aligned masks, alleviating annotation burdens and boosting SAM segmentation IoU across three city test sets by 9.60%, 9.13%, and 20.55% using only 10 real samples.
- Designed a physical-guided SAM decoder for urban PV segmentation to eliminate shadow and reflection interferences, achieving up to a +6.76% gain in Precision and a +10.86% gain in IoU over baseline.

Diffusion-Based Meteorological Super-Resolution for PV Estimation
Supervised by Prof. Zhiling Guo at DigiEnergy Lab. *Oct. 2023 – Sept. 2024*

- Developed a score-based diffusion framework for multimodal meteorological data super-resolution, enhancing spatial resolution from 16 km to 2 km for microclimate-level solar modeling and PV potential estimation.

- Designed SwinT-DyS module by fusing shifted-window self-attention with dynamic grid upsampling to effectively capture fine-grained spatial microclimate patterns.
- Surpassed baselines with a 10.26% MAE reduction, a 1.37% PSNR gain, and a 7.41% SSIM boost, enabling accurate rooftop PV potential mapping.

3D Human Pose Reconstruction System
Supervised by Prof. Liguo Zhang at Cyber-Physical Network System Lab. *Sept. 2023 – May 2024*

- Designed an edge 3D reconstruction system integrating real-time YOLOv8 pose estimation with physical geometric priors, leveraging dynamic spatial filtering to eliminate self-occlusion distortion and background noise in complex environments.

Publications

Jiaze Li, Zhiling Guo, Huan Zhao, Hongjun Tan, Qing Yu, Rui Zhang, Jian Xu, Jinyue Yan. (2024). "Enhancing Multimodal Meteorological Data Resolution via Diffusion Model for Accurate PV Potential Estimation." **International Conference on Applied Energy (Oral)**. https://doi.org/10.46855/energy-proceedings-11399

Quan Hao, Rui Shi, **Jiaze Li**, Liguo Zhang. (2026). "Generative Approach for Detecting Small Intrusive Foreign Objects in High-Speed Railway Scenario." **IEEE Transactions on Intelligent Transportation Systems (IF = 8.4)**, 27(1), 1471–1484. https://doi.org/10.1109/TITS.2025.3625181

Hongjun Tan, Zhiling Guo, **Jiaze Li**, Yuntian Chen, Qi Chen, Haoran Zhang, Jinyue Yan. (2026). "Synthesizing Images with Aligned Masks Using Text-to-Image Based Generative AI for Robust PV Segmentation." **Renewable Energy (IF = 9.1)**, 260, 125217. https://doi.org/10.1016/j.renene.2026.125217

Ruizhe Deng, Zhiling Guo, Penglei Zhang, **Jiaze Li**, Xin Xu, Qi Chen, Yuntian Chen, Jinyue Yan. (2026). "PANEL: A Photovoltaic-Specific Vision-Language Model for Zero-Shot and Few-Shot PV Interpretation Tasks in Remote Sensing." **ISPRS Journal of Photogrammetry and Remote Sensing (IF = 12.2)**, 238, 337–355. https://doi.org/10.1016/j.isprsjprs.2026.05.012

Lingchengjia Zhou, Kechuan Dong, Hongjun Tan, **Jiaze Li**, Qing Yu, Zhiling Guo, Jinyue Yan. (2024). "A High-Precision Method for Photovoltaic Panel Segmentation Combining Large-Scale Model Prior Knowledge and Multimodal Information." **Applied Energy Symposium and Forum: Low-Carbon Cities and Urban Energy Systems (Oral)**. https://doi.org/10.46855/energy-proceedings-11313

Heng Deng, **Jiaze Li**, Quan Hao, Zhaoyang Cheng, Rui Shi, Liguo Zhang. (2024). "Real-Time 3D Human Pose Reconstruction Based on Depth Camera and YOLOv8 Model." ***CN Patent***, CN118823228A.

Academic Service

- Reviewer for *IEEE Transactions on Intelligent Transportation Systems*

Selected Awards

- **Outstanding Student Foundation Scholarship**, Beijing University of Technology *2025*
- **Innovation and Entrepreneurship Award**, Beijing University of Technology *2024, 2025*
- **Academic Excellence Award**, Beijing-Dublin International College *2022, 2023, 2024, 2025*

Skills

- **Languages:** Chinese (native), English (TOEFL iBT 101/120), Japanese (JLPT N2)
- **Programming & Libraries:** Python, C, Java, Bash, PyTorch, MMCV, OpenCV, Diffusers, NumPy, Pandas
- **Tools:** LaTeX, Markdown, WandB, Linux, Git, Anaconda